

Dynamic Partnership with Dissolution Risk

A Preliminary Analysis

September 15, 2026

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Dissolution: divorce.
- **Professional partnership** (law firms, medical practices, consultancies):
a client base, reputation, shared expertise.
Dissolution: a partner leaves or the practice splits.
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■ How do partners invest in joint capital over time?

■ How to design institutions governing dissolution for efficient investment?

Model

First-Best Problem

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- While together: joint flow $F(X_t)$, **split equally** ($F' > 0$, $F'' \leq 0$)
- The partnership ends at Poisson rate λ

Nothing is reversible: $\dot{X}_t = a_t + b_t \geq 0$.

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(d captures how institutions protect individual contributions; d can be negative.)
- Symmetric Nash bargaining:

$$u^A = d(A) + \frac{1}{2}\{S(X) - d(A) - d(B)\} = \frac{1}{2}S(X) + \frac{1}{2}(d(A) - d(B)).$$

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$$\max_{n: \mathbb{R}_+ \rightarrow [0,2]} \int_0^\infty e^{-(r+\lambda)t} [F(X_t) + \lambda S(X_t) - c n_t] dt,$$

subject to $\dot{X}_t = n_t$. (n_t denotes the joint investment $a_t + b_t$ at time t)

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Regularity Assumption

Define $M(X) := F'(X) + \lambda S'(X)$.

1. $M'(X) < 0$ for all $X \geq 0$;
($F(X) + \lambda S(X)$ is strictly concave; at least one of them is nonlinear)
2. $M(0^+) > (r + \lambda)c > \lim_{X \rightarrow \infty} M(X)$.

Efficient Investment: First-Best Characterization

Theorem

Under the regularity assumption, the optimal policy is bang-bang: invest at the maximum rate while $X_t < X^{FB}$ and stop thereafter, i.e.,

$$n_t^{FB} = \begin{cases} 2, & \text{if } X_t < X^{FB}, \\ 0, & \text{if } X_t > X^{FB}, \end{cases}$$

where

$$M(X^{FB}) = (r + \lambda) c.$$

HJB equation: Heuristic Derivation

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Over $[t, t + dt]$ the flow $F - cn$ accrues; with probability λdt the relationship *ends* and the pair collects the *lump* $S(X_t)$; otherwise it continues and is worth $W(X_{t+dt})$.

$$W(X_t) = \max_{n_t \in [0,2]} \left\{ [F(X_t) - c n_t] dt + \underbrace{\lambda dt S(X_t)}_{\text{breakup}} + \underbrace{(1 - (r + \lambda)dt) W(X_{t+dt})}_{\text{survival}} \right\} + o(dt)$$

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Using $W(X_{t+dt}) = W(X_t) + W'(X_t) n_t dt$, cancelling $W(X_t)$ and dividing by dt :

$$(r + \lambda) W(X) = \max_{n \in [0,2]} \{ F(X) + \lambda S(X) + n(W'(X) - c) \}$$

Optimal policy

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$$W(X) = \frac{F(X) + \lambda S(X)}{r + \lambda} \quad \implies \quad W'(X) = \frac{M(X)}{r + \lambda}.$$

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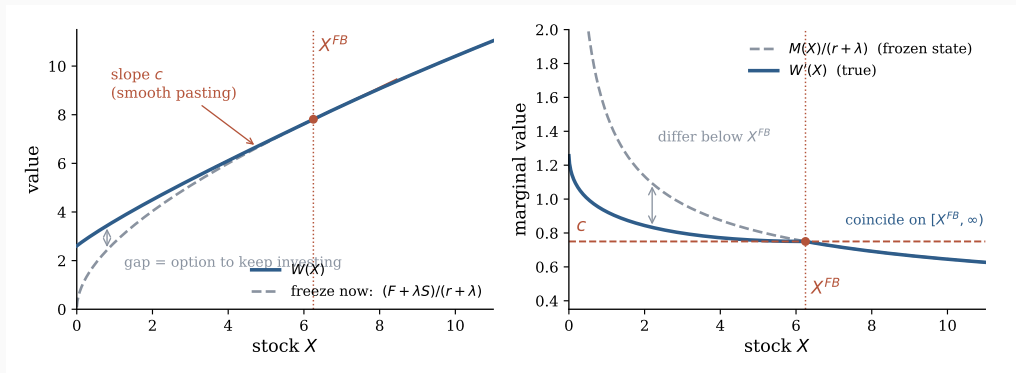
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- Below X^{FB} , $M(X) > (r + \lambda)c$, so $W'(X) > c$ and $n = 2$. At the boundary $W'(X^{FB}) = c$: **smooth pasting**.

Optimal policy: the value function

$$F(X) = \sqrt{X}, S(X) = X; r = 0.3, \lambda = 0.1, c = 0.75 \Rightarrow X^{FB} = 6.25$$



Left: W lies above the freeze-now value below X^{FB} and coincides with it above, meeting tangentially with slope c . Right: the frozen-state slope is *not* W' below X^{FB} ; they meet exactly at the threshold.

Equilibrium

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- **Strategies.** Markov: a single function $\alpha(\text{own stock, other's stock}) \in [0, 1]$, so

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- **Equilibrium.** A *symmetric Markov perfect equilibrium* is an α that is a best response to itself.

Ava's problem

Ava takes α — Bob's strategy — as given and solves

$$\max_{a: \mathbb{R}_+ \rightarrow [0,1]} \mathbb{E} \left[\int_0^\tau e^{-rt} \left(\frac{1}{2} F(A_t + B_t) - c a_t \right) dt + e^{-r\tau} u^A(A_\tau, B_\tau) \right],$$

subject to $\dot{A}_t = a_t$, $\dot{B}_t = \alpha(B_t, A_t)$ and $u^A = \frac{1}{2}S(X) + \frac{1}{2}(d(A) - d(B))$.

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The same Poisson step as in the planner's problem gives

$$\max_a \int_0^\infty e^{-(r+\lambda)t} \left[\frac{1}{2} F(A_t + B_t) + \lambda u^A(A_t, B_t) - c a_t \right] dt.$$

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■ Two changes from the planner: Ava gets $\frac{1}{2}F$ rather than F , and u^A rather than S .

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$$V(A, B) = \max_{a \in [0, 1]} \left\{ \left[\frac{1}{2} F(X) - c a \right] dt + \underbrace{\lambda dt u^A}_{\text{breakup}} + \underbrace{(1 - (r + \lambda) dt) V(A + a dt, B + b dt)}_{\text{survival}} \right\} + o(dt)$$

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Using $V(A + a dt, B + b dt) = V + V_A a dt + V_B b dt$ and dividing by dt :

$$(r + \lambda)V = \max_{a \in [0,1]} \left\{ \frac{1}{2} F(X) + \lambda u^A + \underbrace{b V_B}_{\text{Bob builds}} + a(V_A - c) \right\}$$

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$(r + \lambda)V = \frac{1}{2}F(X) + \lambda u^A(A, B)$ and hence

$$V_A(A, B) = \frac{1}{2} \frac{F'(X) + \lambda S'(X) + \lambda d'(A)}{r + \lambda} = \frac{1}{2} \frac{M(X) + \lambda d'(A)}{r + \lambda}.$$

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- On the symmetric path $A = B = X/2$, define the **private marginal value**

$$m(X) := \frac{1}{2}M(X) + \frac{\lambda}{2}d'(X/2).$$

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■ Ava stops $\iff m(X) \leq (r + \lambda)c$. Compare $M(X) \leq (r + \lambda)c$ for the planner.

Equilibrium Characterization

Regularity Assumption 2

1. $m'(X) < 0$ for all $X \geq 0$;
($M' < 0$ and $d'' \leq 0$ imply this since $m(X) = \frac{1}{2}M(X) + \frac{\lambda}{2}d'(X/2)$)
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2. $m(0^+) > (r + \lambda)c > \lim_{X \rightarrow \infty} m(X)$.

Theorem

Under Regularity Assumption 2, in any symmetric Markov perfect equilibrium, accumulation stops at X^* given by

$$m(X^*) = (r + \lambda)c, \quad m(X) = \frac{1}{2}M(X) + \frac{\lambda}{2}d'(X/2).$$

Comparison with the first best

Planner $M(X^{FB}) = (r + \lambda)c, \quad M(X) = F'(X) + \lambda S'(X)$

Equilibrium $m(X^*) = (r + \lambda)c, \quad m(X) = \frac{1}{2}M(X) + \frac{\lambda}{2}d'(X/2)$

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Theorem

Suppose Regularity Assumption 1 and $d'' \leq 0$ hold.

$$X^* \leq X^{FB} \quad \Longleftrightarrow \quad \lambda d'(X^{FB}/2) \leq (r + \lambda)c$$

Proof: Draw a graph

Can the division rule restore efficiency?

Corollary (efficient division rule)

$X^* = X^{FB}$ if and only if

$$d'\left(\frac{X^{FB}}{2}\right) = \frac{(r + \lambda)c}{\lambda} = \frac{F'(X^{FB})}{\lambda} + S'(X^{FB}).$$

But d cannot be that steep for free

Recall **feasibility**: $d(A) + d(B) \leq S(A + B)$. At the efficient symmetric split $A = B = X^{FB}/2$ this reads $2 d(X^{FB}/2) \leq S(X^{FB})$.

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Proposition (efficiency requires losses in disagreement)

Let d be concave and feasible. If $X^* = X^{FB}$ then

$$d(0) \leq \frac{X^{FB}}{2} \left[\frac{S(X^{FB})}{X^{FB}} - \frac{(r + \lambda)c}{\lambda} \right].$$

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Proof: concavity gives $d(0) \leq d(A) - A d'(A)$ at $A = X^{FB}/2$; feasibility bounds $d(A)$ by $S(X^{FB})/2$; the Proposition replaces $d'(A)$ by $(r + \lambda)c/\lambda$.

Linear dissolution value: $S(X) = kX$

Recall that at the efficient stock, using $F' > 0$,

$$(r + \lambda)c = F'(X^{FB}) + \lambda S'(X^{FB}) = F'(X^{FB}) + \lambda k \quad \implies \quad \frac{(r + \lambda)c}{\lambda} > k$$

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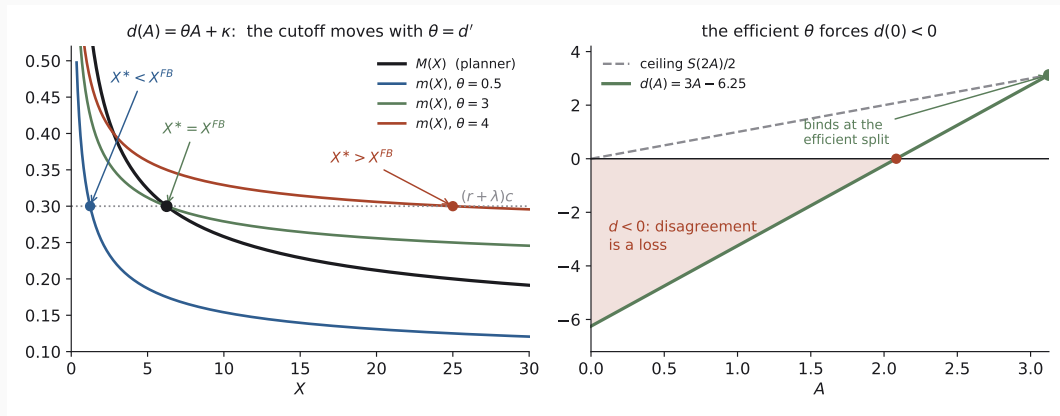
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- Reaching the efficient stock requires negative disagreement payoffs for sufficiently low contributions.
- If disagreement payoffs must be nonnegative, partners underinvest.

Comparison: a numerical example

$$F(X) = \sqrt{X}, S(X) = X, d(A) = \theta A + \kappa; r = 0.3, \lambda = 0.1, c = 0.75 \Rightarrow X^{FB} = 6.25$$



Efficiency needs $\theta = (r + \lambda)c/\lambda = 3$, against a feasibility ceiling of $k = 1$ for proportional rules.

Binding feasibility at the efficient split gives $\kappa = \frac{1}{2}[S(X^{FB}) - \theta X^{FB}] = -6.25$:

Future Directions

- **Endogenous dissolution**

When do partners choose to split, and how do dissolution rules affect that choice?

- **Joint investment versus outside opportunities**

Do institutions encourage building together or preparing to leave?

- **Unequal investment costs**

How should dissolution rights be allocated when one partner can contribute more cheaply?